

Indian Institute of Information Technology Kottayam

Department of Electronics and Communication Engineering

in association with

Next Generation Communications Research Group
organizes

Five Day FDP cum Hands-on Workshop
on

Wireless Communication Technologies for Beyond 5G Systems

(Hybrid Mode)
24 – 28 June, 2024



Indian Institute of
Information Technology
Kottayam



Organizing Committee

Patrons

Dr. Rajiv V. Dharaskar, Director, IIITK.
Dr. M. Radhakrishnan, Registrar, IIITK.
Dr. P. Mohanan, PIC Academics, IIITK.

Convenor

Dr. A. Ananth,
Assistant Professor and Head,
Department of ECE, IIITK
☎ 0482 220-2176
✉ ananth@iiitkottayam.ac.in

Coordinators

Dr. Rajesh G, Assistant Professor, IIITK
☎ 9400321566
✉ grajesh@iiitkottayam.ac.in
Dr. Vineeth P, Assistant Professor, IIITK
☎ 9787767640
✉ vineeth@iiitkottayam.ac.in

Registration

Registration Link:
<https://forms.gle/JYzwv4ntjwXSFxAN9>

Deadline: 22nd June 2024



Indian Institute of Information Technology Kottayam, Valavoor P.O., Pala, Kottayam- 686635, Kerala.

About IIIT Kottayam

Indian Institute of Information Technology Kottayam at Valavoor, Pala, Kerala is one among the IIITs that have been established as "Institution of National Importance" by MoE, Govt. of India under the ambit of IIIT PPP act 2017. IIIT Kottayam campus is situated at Valavoor, Pala, Kottayam - a picturesque location, spread over in 53 acres of land - allotted by Govt. of Kerala in the southern part of the Country commonly referred to as God's Own Country. B.Tech students are admitted through the JoSAA/CSAB counselling against the JEE main rank list. Institute offers 4 year B.Tech and B.Tech (Hons) programs in ECE, CSE, CSE with specialization in AI & Data Science and Cyber Security. In addition, Institute also offers Ph.D. programs in ECE, CSE and Computational Mathematics as well as M.Tech programme in (1) AI & Data Science (2) Cyber Security (3) Big Data & ML for the working professionals. Further, IIIT Kottayam offers Integrated M.Tech in AI & Data Science for working professionals.



About the Department

Electronics and Communication Engineering Department of IIIT Kottayam adopts foremost teaching and learning practices for the students to excel in theoretical and experimental learning activities. The department paves a way to create highly motivated and technologically competent human resource in various domain through effective partnership with research organizations, industries, and active communities, across the globe. Our B.Tech. program in ECE enhances intellectual and practical skills of the students, by providing exposure to state-of-the-art technologies and moulds the students with fundamental competencies, problem solving abilities and entrepreneurship skills to excel in their career.

Our curriculum imparts strong foundation on Electronics and Communication Engineering with a blend of Computer Science, extending to elective courses in emerging fields. This program spans across prominent fields including Hardware/Software Co-Design, Signal Processing, VLSI and Embedded Systems, IoT, Edge computing, Next Generation Communication Network, Integrated Circuits, Robotics and Automation, Artificial Intelligence, Machine Learning, Deep Learning and so on. This dynamic curriculum acquaints the students for current industrial needs as well as to pursue higher studies.

Speakers

1. Dr. Ananth A, Assistant Professor, Dept. of ECE, IIIT Kottayam
2. Dr. Deepak Jose, Assistant Professor, Dept. of CSE, IIIT Kottayam
3. Dr. Emy Mariam George, Assistant Professor, Dept. of CSE, IIIT Kottayam
4. Dr. Rajesh G, Assistant Professor, Dept. of ECE, IIIT Kottayam
5. Dr. Vineeth Palliyembil, Assistant Professor, Dept. of ECE, IIIT Kottayam

A few other experts from academia (IITs, NITs etc.) will also be delivering talks and provide hands-on sessions.

Participants

Targeted participants are UG, PG students, Research Scholars, Faculty members of academic institutions and scientists/researchers from industries and research organizations.

Important information to the participants:

- Accommodation will be provided in the institute hostels for the offline participants.
Single room: Rs. 200/- per day
Shared room: Rs. 100/- per day
- Working lunch and tea will be provided to the offline participants.

Registration Fee:

Faculty/ Industry Professional: Rs. 1500/-
UG/PG students, Research Scholars: Rs. 1000/-

Payment Link:

<https://www.onlinesbi.sbi/sbicollect/>

1. Select Category "Educational Institutions"
2. Search for Payee "IIIT Kottayam" and click on it
3. Payment Category – Select "NGCRG"
4. Enter the details and make payment
5. Use the DU number of fee payment to complete the form

Note: Registration fee is non-refundable.

About the Workshop

This five day FDP cum hands-on workshop will introduce various communication technologies like simulation and performance evaluation of digital modulation techniques over AWGN and Rayleigh fading channels, Index Modulation, NOMA, RIS, VLC, OWC, RF/FSO, WBAN. Further, the sessions shall demonstrate the implementation aspects of the above-mentioned wireless communication techniques through MATLAB/open source tools like Python for better understanding.

The main objectives of the workshop are:

- To understand the aspects of index modulation, a promising technique for next-generation wireless communication systems.
- To understand non-orthogonal multiple access (NOMA) radio access techniques in 5G communication scenarios such as M2M and IoT.
- To know about the potential use cases of reconfigurable intelligent surfaces (RIS), visible light communication (VLC), optical wireless communication (OWC), RF/FSO in 5G and beyond wireless networks.
- To understand the recent developments and technological advancements in wireless body area networks (WBAN).
- To explore the research opportunities and challenges in the field of 5G and beyond wireless communication, fostering discussions on emerging trends and future directions.

